Electrochimica Acta 56 (2011) 3079–3084

Contents lists available at ScienceDirect

Electrochimica Acta

journal homepage: www.elsevier.com/locate/electacta

Studies on surface preparation and smoothness of nanostructured Bi_2Te_3-based alloys by electrochemical and mechanical methods

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ARTICLE INFO

Article history:
Received 20 August 2010
Received in revised form 4 December 2010
Accepted 4 December 2010
Available online 16 December 2010

Keywords:
Nanostructures
Bismuth telluride based alloys
Thermoelectrics
Electrolysis
Chemical mechanical polishing

ABSTRACT

Significant improvements in the dimensionless thermoelectric figure-of-merit (ZT) for nanostructured bismuth telluride, Bi_2Te_3, and its alloys have been demonstrated. In designing high-performance thermoelectric devices, variations in the thermal and electrical contact resistances due to interfacial effects between the nanostructured alloy and the metallic electrodes remain a significant issue. Smooth scratch-free surfaces should provide a baseline for contact resistance studies. In this paper, the root mean square roughness over a 10 μm^2 of nanostructured bismuth tellurium based alloys was reduced from 133 nm to 1.9 nm by a procedure consisting of electrolysis, mechanical polishing, and chemical mechanical polishing (CMP). Post-CMP cleaning was also developed to yield a wettable surface for the subsequent conformable metallization.

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1. Introduction

Nanostructured thermoelectric alloys exhibiting an enhanced dimensionless thermoelectric figure-of-merit (ZT) have attracted considerable interest recently for use in high-performance thermoelectric devices [1–3]. In particular, nanostructured p-type bismuth antimony tellurides have achieved a peak ZT of about 1.4 at 100 °C [1]. Bi_2Te_3-based alloys ($Bi_xSb_{2-x}Te_3$ as the p-type and $Bi_2Se_yTe_{3-y}$ as the n-type) are the most widely used thermoelectric materials near room temperature. Device performance not only depends on ZT, but also on the electrical and thermal contact resistances between the electrodes and the thermoelectric materials [4,5]. Interfacial electrical and thermal contact resistances depend on the interface morphology. Surface roughness is an important parameter to control the interface morphology. One limit of the interface control is to achieve smooth and scratch-free surfaces. However, very little research has been conducted in the surface treatment of Bi_2Te_3-based alloys, especially nanostructured thermoelectric materials [6]. In conventional polishing processes, scratches are inevitably generated on the surface because of the relatively low hardness of Bi_2Te_3-based alloys compared with abrasives such as alumina or silica. In addition, nanostructured alloys have many grain boundaries which are weak points for mechanical or chemical treatment.

This article presents a technique to prepare smooth scratch-free surfaces of nanostructured Bi_2Te_3-based alloys by electrolysis, mechanical polishing, and chemical mechanical polishing (CMP). A post-CMP method was also developed to achieve a wettable surface for the subsequent metallization. Root mean square (RMS) surface roughness of 1.9 nm is achieved. Such smooth surfaces will be of fundamental interest for studying interfacial electrical and thermal contact resistances, and is particularly convenient for interfacial thermal resistance measurements using time-domain thermoreflectance (TDTF) technique [7–9].

2. Experimental

2.1. Sample fabrication

Bulk nanostructured Bi_2Te_3-based materials, p-type (alloyed with Sb as $Bi_xSb_{2-x}Te_3$) and n-type (alloyed with Se as $Bi_2Se_yTe_{3-y}$) disk samples of 25 mm in diameter and 2 mm in thickness were made by a ball milling and hot pressing method [1]. After hot pressing, both sides of the disk sample were polished using sand paper. This initial surface was used as a reference for the polishing process. A South Bay Technologies Model 920 polishing tool was used for all polishing processes.

2.2. Measurements

A Nanoscope IIIa atomic force microscope (AFM) was used to scan the surfaces and calculate the root mean square (RMS) rough-

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doi:10.1016/j.electacta.2010.12.008

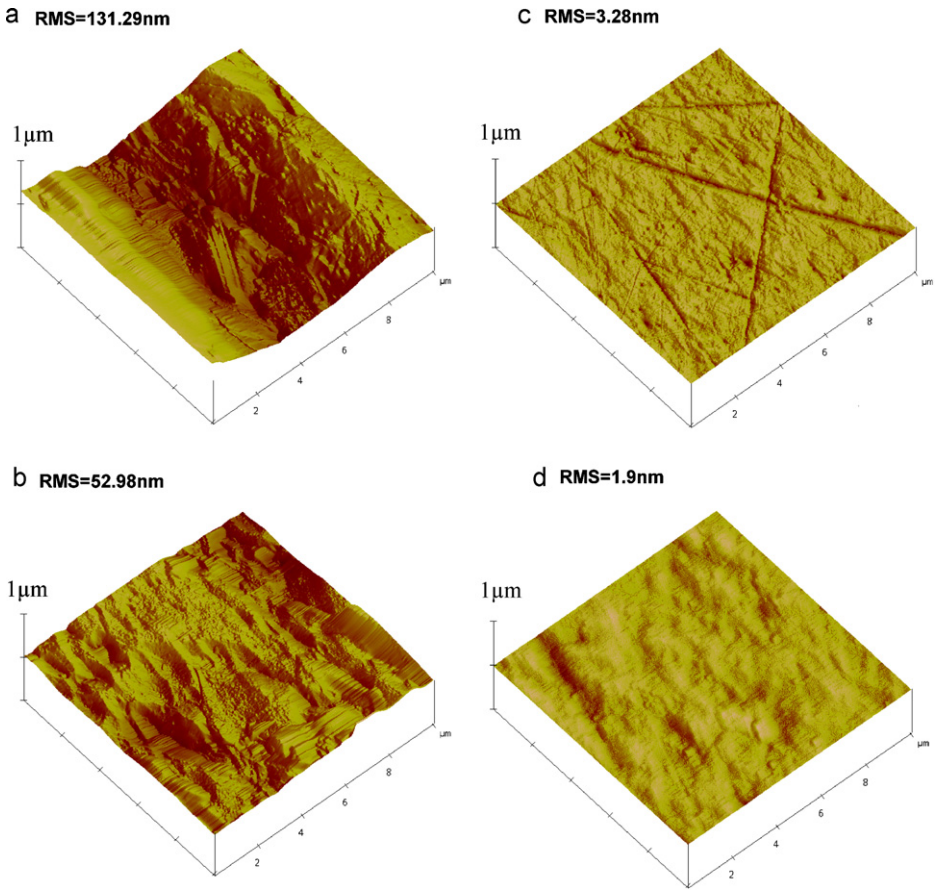


Fig. 1. Atomic force microscopy images of surfaces of nanostructured $Bi_2Se_yTe_{3-y}$ alloys, (a) as-made, (b) after electrolysis, (c) after mechanical polishing, and (d) after chemical mechanical polishing. Scan size is 10 μm, scan rate is 1 Hz. The *Z* axes for the height is 1 μm/div and *X* axes is 2 μm/div.

ness. A KLA-Tencor P16 profilometer was used to do line-scan across the surface for dissolution rate evaluation. Contact angle was measured by a static contact angle goniometer, Kyowa Kaimen CA-X. Electrochemical measurement, including cyclic voltammogram, open circuit potential, static potentiodynamic polarization was performed using a Gamry PCI4/300 potentistat/galvanostat system. A platinum mesh was used as the counter electrode and a saturated calomel electrode (SCE) was used as the reference electrode in standard three-electrode system. The experiments were successively carried out on n-type nanostructured $Bi_2Se_yTe_{3-y}$ alloys at first and then on p-type $Bi_xSb_{2-x}Te_3$.

3. Results and discussion

Fig. 1(a) is an AFM image of the initial surface and exhibits a root mean square roughness of 133 nm. At this roughness, mechanical polishing will require a long processing time to reach a low surface roughness. However, polishing for an extended period of time has a high possibility of generating deep scratches or chemical corrosion, particularly in nanostructured alloys which are prone to such issues due to the abundant grain boundaries. The details of Figs. 1(b)–(d) are described in the subsequent sections.

A previous paper reported that mixtures of NaOH–tartaric acid form a good electrolyte for electrolytic etching of Bi_2Te_3 with the goal of achieving a preliminary flat surface to reduce processing time for the following polishing steps [6]. However, very few papers provided detailed explanation of this method. Here, we tried to understand and utilize this method from an electrochemical perspective. We were interested in the range of pH and potentials where it is possible to dissolve both major species of bismuth and tellurium in an aqueous solution. According to the Pourbaix diagram of Bi_2Te_3, tellurium is soluble as TeO_4^{2-} but bismuth precipitates as insoluble Bi_2O_3 in a range at pH > 7.5 [10,11]. If applying potentials larger than 1.5 V at pH > 10.5, tellurium is still soluble as TeO_4^{2-} and bismuth would form Bi_2O_5. Based on reactions relating aqueous bismuth and tellurium species and its precipitates, Bi_2O_5 would react with hydrogen cations to become Bi^{3+} which would dissolve into solution [11,12].

Fig. 2 is the anodic wave of cyclic voltammetry in a 10% sodium hydroxide solution and a mixture of 10% sodium hydroxide and 5%

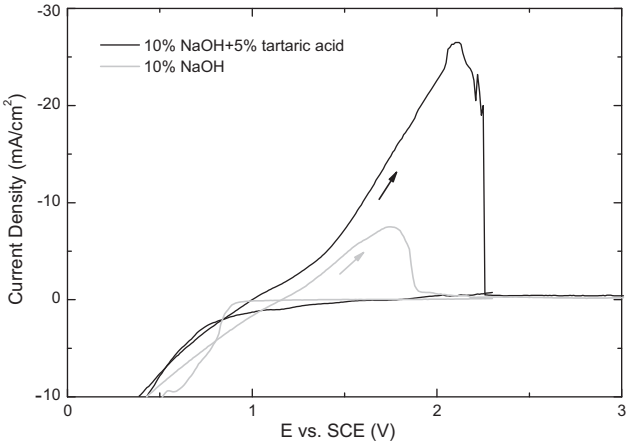


Fig. 2. Anodic wave of cyclic voltammetry in a 10% sodium hydroxide solution and a mixture of 10% sodium hydroxide and 5% tartaric acid.

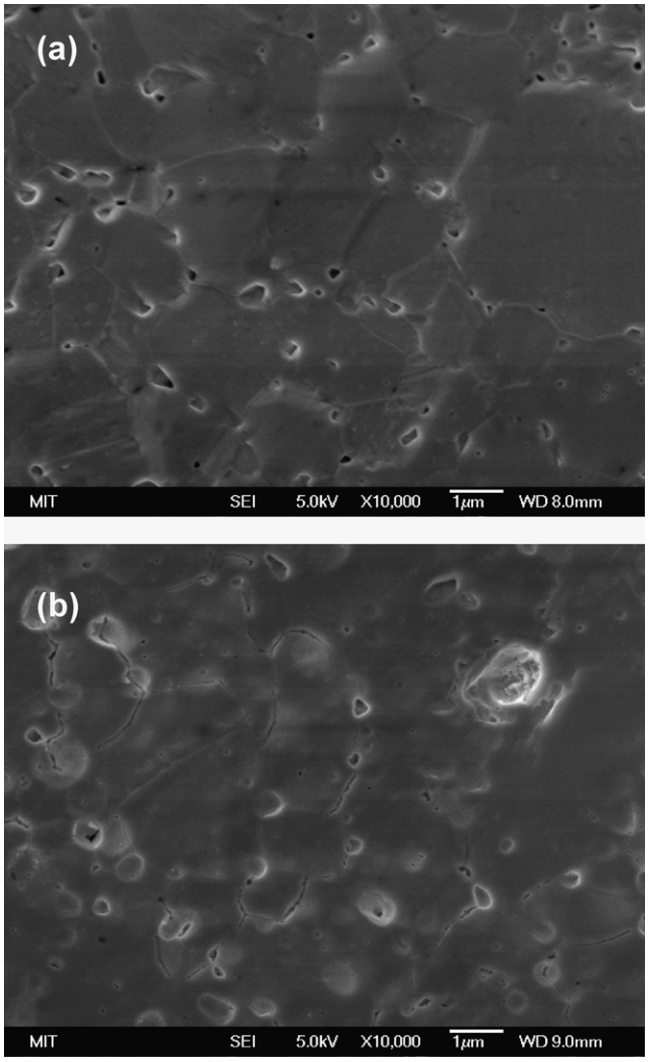


Fig. 3. SEM images of nanostructured $Bi_2Se_yTe_{3-y}$ alloys after electrolysis, (a) 30 s and (b) 60 s.

tartaric acid to make pH 11.5 solutions. The rise in current density is less pronounced when tartaric acid is not used. It leads us to believe that tartaric acid performs the role of providing hydrogen cations for Bi_2O_5 in the electrolysis reaction. In principle, an ideal electrolysis process can achieve a low surface roughness by itself, but at the cost of greater surface damage due to a longer processing time. SEM pictures of surfaces after electrolyzing at 1.5 V for 30 s and 1.5 V for 60 s are shown in Fig. 3. These figures confirm that electrolysis at longer processing time damages the surface, with cracks forming at the grain boundaries. Thus, the initial surface was electrolyzed at 1.5 V for 30 s in a mixture of 10% NaOH and 5% tartaric acid to achieve a preliminary flat surface. The RMS roughness was reduced from 133 nm to 52 nm, as seen in Fig. 1(b).

The samples were then mechanically polished on a Buehler Trident cloth pad with an upper soft cloth surface and lower hard polyurethane support. A composite polishing pad is important for nanostructured Bi_2Te_3-based alloys because the upper soft cloth will prevent deep scratches while the lower hard supporting pad will prevent the surface from being polished along its beginning topography. The slurry used here is a 5 wt% 0.05 μm alumina suspension, also commercially available from Buehler. The polishing down-force was 1.1 psi with a counter-clockwise plate rotational speed of 110 rpm and a counter-clockwise carrier rotation of 30 rpm. The slurry flow rate was fixed at 45 mL/min. The low down-force polishing is used to eliminate surface scratches but will extend polishing time. If chemical mechanical polishing was directly used instead, frictional heat would accelerate the chemical reaction, causing corrosion on the nanostructured Bi_2Te_3 surface. For that reason, mechanical polishing was performed prior to chemical mechanical polishing to make the chemical reaction short. The 52 nm RMS roughness of the sample after electrolysis was reduced to 3.2 nm over a 10 μm^2 area after polishing for 30 min, as shown in Fig. 1(c). But, mechanical polishing inevitably generated scratches on the surface of nanostructured Bi_2Te_3-based alloys.

Following the mechanical polishing step, chemical mechanical polishing (CMP) was employed to erase surface scratches. The abrasive used in this slurry consisted of a 5 wt% 0.05 μm colloid silica. The CMP slurry chemistry determines the final polishing performance where the key is to form a passivation layer in order to reduce the isotropic etching of recessed regions on an uneven surface. According to the Pourbaix diagram, Bi_2Te_3 can be passivated in acidic media with pH ranging from 3.5 to 6 [11–13]. Nitric acid, which is a good etchant for $Bi_2Se_yTe_{3-y}$ alloys, was added to adjust the pH. Fig. 4(a) shows the open circuit potential (OCP) of n-type nanostructured $Bi_2Se_yTe_{3-y}$ alloys as a function of the pH value. At pH values ranging from 3.5 to 9, bismuth forms insoluble Bi_2O_3. At a pH of 5, OCP decreased to a stable value indicating the formation of passivation layer. At a pH of 3.5, OCP decreased first and then increased. The reason could be that TeO_3 is not stable at low pH acidic solution and would be reduced to TeO_2 or $HTeO_2^+$ based on thermodynamic expressions of the Bi_2Te_3 system [10,12]. For general conditions, it would form $HTeO_4^-$ dissolved in slurry to increase OCP. Therefore, the pH needs to be controlled between 4 and 7 to prevent Te dissolution. So, the slurry pH we used here was 5. In general, hydrogen peroxide (H_2O_2) can be utilized in polishing slurries as an oxidizer to enhance the passivating reaction and some inhibitors, such as benzotriazole (BTA), are also added to avoid isotropic etching. Fig. 4(b) is the polarization scans with and without H_2O_2 and BTA. At a pH of 5, there is a sign of passivation at the anodic current branch. The film may not completely passivate the surface but may still decrease the dissolution rate, as is expected according to the results for open circuit potential. In the presence of 1 vol% hydrogen peroxide, the corrosion potential increased and the corrosion current density decreased. This result reveals that the effect of adding an extra oxidizer would decrease the corrosion rate. We tried to add more hydrogen peroxide up to 5 vol% into the system, but the corrosion potential and current did not show any significant change. The reason could be that the surface is already in its passive state above 1 vol% H_2O_2 and that etching is likely controlled by ionic diffusion through a passivation layer [13].

In an efficient CMP operation, the dissolution rate at a recessed region on the uneven surface should be as small as possible to avoid isotropic etching. Benzotriazole (BTA) is a well-known corrosion inhibitor for copper, copper-alloys, and other metal surfaces [14]. The effect of BTA on metal CMP to reduce the corrosion in the recessed areas has also been studied [15]. If a 0.1 vol% BTA is added into 1 vol% H_2O_2 pH 5 slurry, the potential drop became more positive and the corrosion current density decreased, and there was pronounced passivation sign between 0 V and 0.8 V SCE (saturated calomel electrode), as shown in Fig. 4(b). Fig. 4(c) shows an average dissolution rate to provide evidence that a protective layer was formed using both 1% hydrogen peroxide and 0.1% benzotriazole in slurry. Based on previous studies [14–17], even a small amount of BTA reduces the dissolution rate significantly and further increase of BTA concentration does not enhance the metal protection. As mentioned above, CMP slurry for the nanostructured $Bi_2Se_yTe_{3-y}$ alloys is 1% hydrogen peroxide, 0.1% benzotriazole, and 5 wt% 0.05 μm colloidal silica at pH of 5 adjusted by nitric

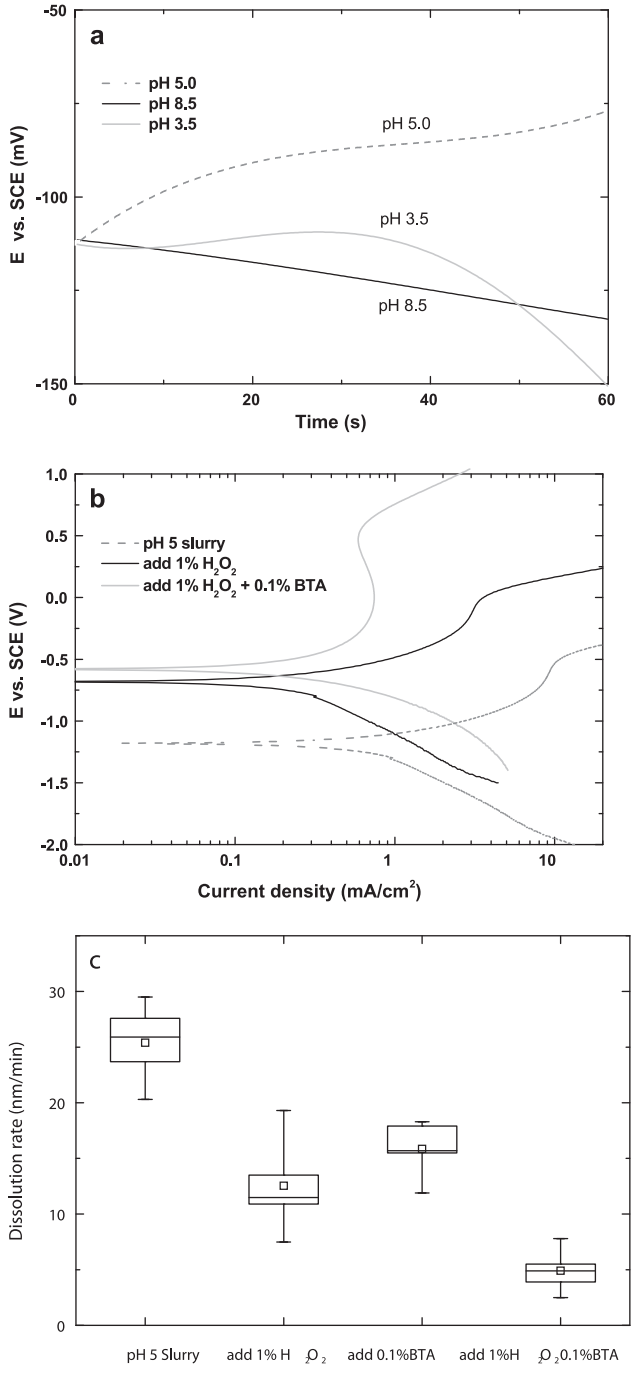


Fig. 4. (a) Variation of OCP with time in the slurry in different pHs, (b) polarization curves for adding H_2O_2 and BTA in pH slurry, and (c) effect of H_2O_2 and BTA on the dissolution rate in pH 5 slurry.

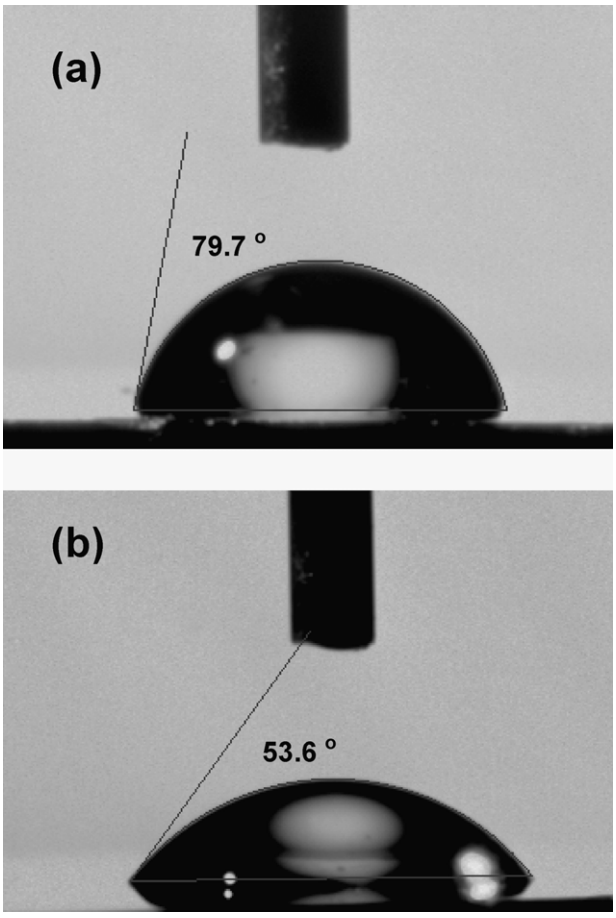


Fig. 5. Contact angles of nanostructured $Bi_2Se_yTe_{3-y}$ alloys (a) before and (b) after post-CMP cleaning.

acid. Fig. 1(d) shows a scratch-free surface after 5 min of chemical mechanical polishing. As a result, a scratch-free surface with 1.9 nm RMS roughness over 10 μm^2 was achieved on nanostructured $Bi_2Se_yTe_{3-y}$ alloy by 30 s electrolysis, 30 min mechanical polishing, and 5 min chemical mechanical polishing.

Since the CMP process leaves abrasive particles, trace metal, and organic component on the surface, post-CMP cleaning is necessary. Any particles left behind has a significant effect in creating imperfections ranging from bumps or pits to regions of excessive contact resistance or to causes non-uniform thin metal transducer layer for pump-and-probe measurement [18,19]. On the other hand, almost the entire surface of $Bi_2Se_yTe_{3-y}$ alloys would be covered with a hydrophobic thin oxide layer after CMP, which functions poorly as nucleation sites for metallization because the growth of metal film on substrates is classified by the same principles as wetting or nonwetting of liquids [20,21]. If the film wets the substrate, the atoms are more strongly bound to the substrate than to each other and grow layer-by-layer (Frank–van der Merwe growth). In contrast, if the film wets poorly, the atoms are more strongly bound to each other than to the substrate and grow three-dimensional islands (Volmer–Weber growth). Volmer–Weber growth is similar to liquid droplets with a large contact angle and is usually not desirable for technological applications. Therefore, post-CMP cleaning must yield a wettable and clean surface beneficial to subsequent metallization processes. The contact angle was used to evaluate wettability and particle removal efficiency [18,22]. Generally, the number of residual slurry particles decreases with decreasing contact angle in the region until reaching a threshold in wettability [22]. A 0.5% bromine in ethanol solution was chosen to remove the thin oxide on Bi_2Te_3-based alloys [6]. After bromine–ethanol treatment, a 5% additive (product name: CLEAN-100, Wako Pure Chemical Industries Ltd.) was chosen to improve the wettability, dissolve the organic components, and promote lift-off of particles from the substrate [19].

After 1 min of immersing in bromine-ethanol at room temperature and a 5 min immersion in CLEAN-100 at 45 °C with sonication, the contact angle with the sample surface and water was measured, as shown in Fig. 5. The sample surface with and without post-CMP cleaning showed a contact angle of 53.6° and 79.7°, respectively. So, it is believable that a metal film would wet on the nanostructured $Bi_2Se_yTe_{3-y}$ surface after cleaning. 100 nm nickel films by sputter-

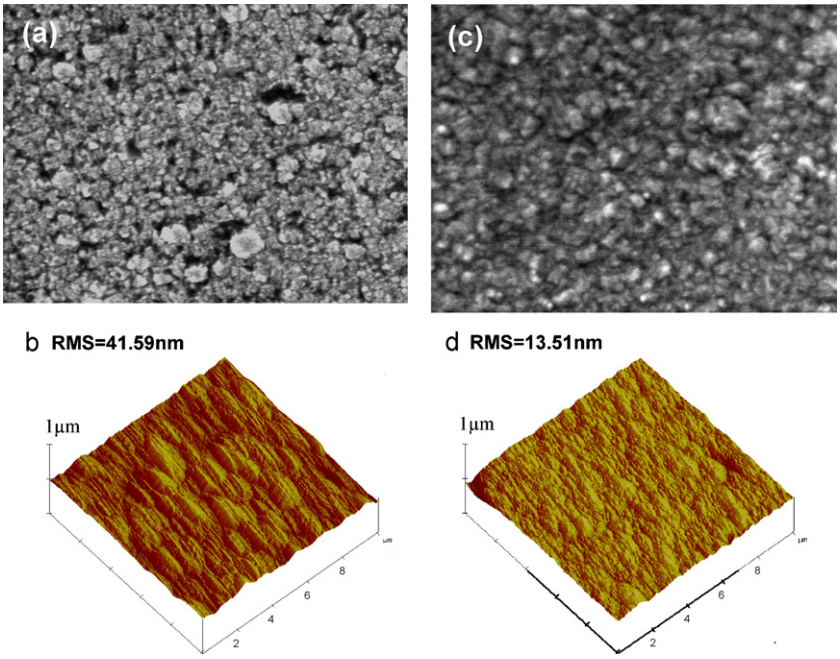


Fig. 6. SEM images and atomic force microscopy images of 100 nm nickel sputtered on the surface of nanostructured $Bi_2Se_yTe_{3-y}$ alloys (a) and (b) without and (c) and (d) with post-CMP cleaning. AFM scan size is 10 μm, scan rate is 1 Hz. The *Z* axes for the height is 1 μm/div and *X* axes is 2 μm/div.

ing on sample surfaces with and without post-CMP cleaning were evaluated by SEM observation (Fig. 6(a) and (c)). The SEM results are consistent with wettability of liquids where a conformable nickel film was formed on the wettable surface, and cracked on the non-wetting surface where metal islands can be clearly seen. AFM was also used to measure the roughness of the 100 nm nickel films sputtered on the surface of nanostructured $Bi_2Se_yTe_{3-y}$ alloys with and without post-CMP cleaning (Fig. 6(b) and (d)). Uniform nickel film can be achieved on wettable surface. Poor wettability has relatively large surface roughness because poor nucleation results in growing three-dimensional islands. These results support the SEM observation and can be explained by Volmer–Weber growth model.

These processes were also successfully reproduced on p-type $Bi_xSb_{2-x}Te_3$ samples with the same performance. We performed this process sequence on several samples of both n- and p-types to prove that the smooth scratch-free surface can be achieved on nanostructured Bi_2Te_3-based alloys. Based on this work, more experiments and investigations can be carried out for a complete understanding of the interfacial effect on mechanical strength and contact resistance of nanostructured Bi_2Te_3-based alloys.

4. Conclusions

In this paper, a new polishing process involving electrolysis, mechanical polishing, chemical mechanical polishing, and post-CMP cleaning was developed to fabricate scratch-free surfaces with a 1.9 nm root mean square roughness over a 10 μm^2 on both n- and p-type nanostructured Bi_2Te_3-based alloys. It was found that an optimal electrolysis treatment can preliminarily reduce surface roughness in a short time. Mechanical polishing, using a composite pad and slurry of 5 wt% 0.05 μm alumina, was then conducted to achieve global planarization. Chemical mechanical polishing was then used to remove scratches by softening the surface. The slurry used in this process consisted of 1% hydrogen peroxide, 0.1% benzotriazole, and 5 wt% 0.05 μm colloidal silica at pH of 5. To achieve a wettable surface for subsequent metallization, a 0.5% bromine in ethanol solution with 5% WAKO CLEAN-100 additive was used for post-CMP cleaning. Conformable metallization by sputtering was then achieved on the wettable and smooth surface.

Acknowledgements

The authors thank Jonathan K. Tong and Dr. Sheng Shen for very helpful discussions and editorial assistance, and also thank Daniel Kraemer for fabricating the polishing jig. This work is supported by "Solid State Solar-Thermal Energy Conversion Center (S^3TEC)", an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under Award Number: DE-SC0001299 (G.C. and Z.F.R.).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.electacta.2010.12.008.

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